

What should we expect before we simulate?

First-principles photon budget and timing optimisation
of a 1.4 m scintillator bar for the SHiP Timing Detector

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Central idea

Geometry + Snell + exponential attenuation predict the dominant trends before any Geant4 output is shown. The estimator choice — how we extract T_0 from photon arrival times — is as important as the reflector or material.

The object: one bar of a 50 m² detector

Bar geometry: 1400 mm $\times$ 60.00 mm $\times$ 10.00 mm

- 8 SiPMs/END face — 16 END total (L+R)
- 20 sparse SiPMs along TOP face
- Vikuiti 3M ESR reflector
- 1 GeV μ^- vertical, through $H = 10$ mm

Best result tested (EJ-230, $x = 0$, optical limit)

$N_{\text{END}} = 16$, $N_{\text{TOP}} = 20$, $N_{\text{total}} = 36$

$\sigma_{\text{END}} = 53.68$ ps $\sigma_{\text{TOP}} = 15.20$ ps $\sigma_{\text{BLUE}} = 15.21$ ps

$w_{\text{TOP}} = 0.987$ BLUE $\approx$ TOP alone.

New in this version

- Estimator definitions made explicit
- Historical ~ 70 ps traced and explained
- BLUE derivation with weights & correlation
- $\sigma(T_0)$ vs position (21% peak-to-peak)
- Pareto frontier: resolution vs channel count
- Role of END: position, not timing

1 — Napkin

Snell, geometry and exponential attenuation fix the order of magnitude and the *direction* of every trend: which material gives more light, which gives better timing.

2 — Geant4

Confirms the scale, adds distributions the napkin cannot produce: Landau fluctuations, path-length dispersion, angular mixing.

Knoll on optical Monte Carlo: these codes are accurate only to the extent that surface reflection, bulk absorption and refractive indices are correct. Those are exactly the inputs a napkin calculation can police.

Bethe–Bloch minimum, through $H = 10.00$ mm

$$\frac{dE}{dx} \approx 2.000 \text{ MeV/cm} \Rightarrow E_{\text{dep}} \approx 2.000 \text{ MeV}$$

- Sets the scale of everything downstream
- Every later factor multiplies this 2.000 MeV
- Width is Landau — simulation territory

What the napkin cannot give

Event-by-event Landau fluctuations, δ -ray production, path-length variation. The mean survives; the width is for Geant4.

$$N_{\gamma} = E_{\text{dep}} \cdot Y \approx 2 \times 9700 \approx 1.9 \times 10^4$$

Optical properties used in simulations:

Material	Yield [ph/MeV]	τ_d [ns]	λ_{att} [cm]
EJ-200	10000	2.1	380
EJ-204	10400	1.8	160
EJ-230	9700	1.5	120

The trade-off

Yields agree to within 7 %. What separates the materials is how much survives 70 cm travel (λ_{att}) and how fast the emission decays (τ_d) — and those two pull in opposite directions.

Critical angle

$$\theta_c = \arcsin(1/n) = 39.27 \text{ deg}, \quad n = 1.580$$

Trapped fraction — slab (Knoll Eq.8.15)

$$a = \cos \theta_c = 0.774, \quad f_{\text{TIR}} = 2a - 1 = 0.5484$$

Both walls: $f_{\text{TIR}}/\text{end} = 0.2742$

The bar is not a slab

Two wall families must hold TIR simultaneously.
Knoll Eq. 8.14 (cylinder, meridional rays) gives 18.4% per direction — different geometry.

Key number

≈ 0.2742 of photons are guided toward each end.

Group velocity

$$v_g = c/n \approx 189.7 \text{ mm/ns}$$

Effective velocity (END, from Geant4)

$$v_{\text{eff}} = 184.6 \text{ mm/ns} \Rightarrow \beta_{\text{eff}} \approx 13.37 \text{ deg}$$

β_{eff} is the effective axial angle of the photon population that dominates the earliest arrival times.

END vs. TOP select different photons

END is dominated by nearly axial, early photons.
TOP integrates markedly more diagonal trajectories.
These two faces carry *different* timing information, motivating the BLUE combination.

Geometric estimate, ray just outside TIR

$$N_{\text{bounce}} \approx \frac{d \tan \alpha_c}{D}, \quad \alpha_c = 50.73 \text{ deg}$$

With $d = 700 \text{ mm}$, $\tan \alpha_c = 1.223$:

Face	D	N_{bounce}
thin (H)	10.00 mm	≈ 86
wide (W)	60.00 mm	≈ 14

Factor of 6 difference

The same photon direction produces $\sim 6\times$ more bounces against the 10 mm face than the 60 mm face. Reflector quality matters more for the thin faces.

Step 6 — Reflector losses become exponential attenuation

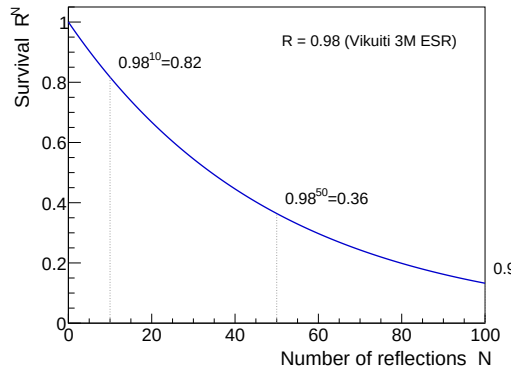
Survival after N bounces

$$P_R = R^N = e^{N \ln R}$$

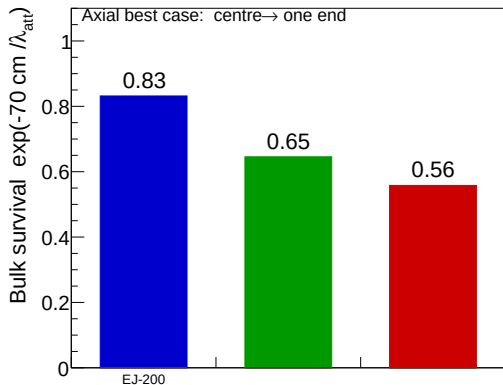
$$\Lambda_{\text{refl}} = \frac{-D}{\tan \alpha_c \cdot \ln R}$$

With $R = 0.98$ (Vikuiti, napkin):

Face	D	Λ_{refl}
thin (H)	10.00 mm	≈ 0.4 m
wide (W)	60.00 mm	≈ 2.4 m



Step 7 — Bulk attenuation and decay time pull in opposite directions



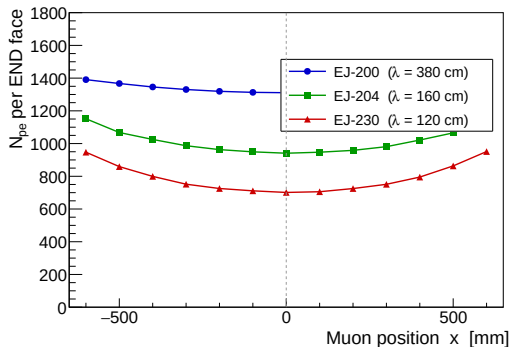
Napkin prediction (before the run)

EJ-200 delivers the most light (largest λ_{att}).
EJ-230 loses the most, but its faster kinetics ($\tau_d = 1.500 \text{ ns}$ vs. 2.100 ns) should win on timing.

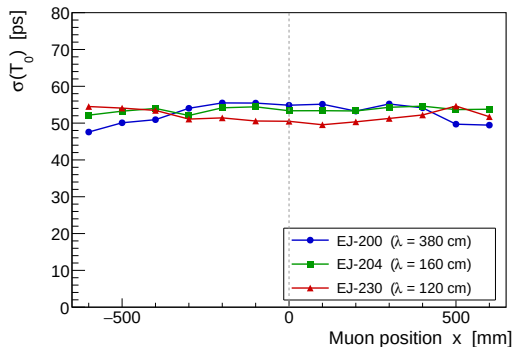
Geant4 confirms both orderings

N_{pe} : EJ-200 > EJ-204 > EJ-230
 $\sigma(T_0)$: EJ-230 < EJ-204 < EJ-200
Both directions as predicted.

The ordering was predicted, then observed



N_{pe}/END vs. position. EJ-200 > EJ-204 > EJ-230, steeper fall for shorter λ_{att} .



$\sigma(T_0)/\text{END}$ vs. position. Ordering inverted: EJ-230 < EJ-204 < EJ-200, driven by τ_d .

Two figures, two orderings, opposite directions — written down before the simulation ran. These are END-face results; TOP SiPMs not yet included.

Closing the chain: photon budget for EJ-230 at $x = 0$

Step	Factor	Cumulative (per end)
Produced	$Y \times E_{\text{dep}} = 9700 \times 2$	$\sim 1.94 \times 10^4$
TIR-guided toward one end	$\times 0.2742$	~ 5320
Bulk survival (70 cm)	$\times 0.5580$	~ 2970
End-face coverage (288.0 mm ² /600.0 mm ²)	$\times 0.4800$	~ 1425
PDE (≈ 0.4000)	$\times 0.4000$	~ 570.0 pe PE/end
Geant4 (EJ-230, END-only, Vikuiti, $x = 0$)	—	≈ 700 PE/end

Scale check

Napkin: ~ 570 PE/end. Geant4: ~ 700 PE/end.
Agreement at 20 % level. The excess comes from the reflector-recovered population (non-TIR paths not counted in the napkin chain).

Count per end — always

Geant4 total $L + R \approx 1400$ PE. Compare with napkin **570 PE/end**, not with 1400 PE. Mixing per-end and $L+R$ silently doubles the answer.

Quantity	Napkin / literature	Verdict
Photons per muon	$\approx 2 \times 10^4$	consistent
Trapped fraction (one wall)	$a = 0.774$, Knoll 8.15	textbook match
Group velocity	≤ 189.7 mm/ns	consistent
N_{pe} ordering	EJ-200 > EJ-204 > EJ-230	confirmed
$\sigma(T_0)$ ordering	EJ-230 < EJ-204 < EJ-200	confirmed
σ_x from $\sigma_{\Delta t}$	10.29 mm	internal check
Napkin PE/end vs. Geant4	570 vs. ~ 700 (+20%)	consistent
Cylinder trapping (EJ-228)	18.4%/dir (Eq. 8.14)	different defn.

7 agreements. 1 definition check. The napkin correctly identifies the reflector-recovered population as the source of the 20% surplus.

Two photon populations carry different information

Population A — TIR-guided

Large $|u_x|$, few bounces, v_x close to c/n , arrives first. The leading-edge (earliest- k) estimator is dominated by this population.

Population B — reflector-recovered

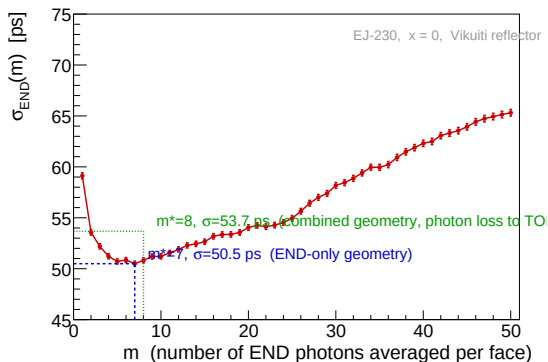
Large transverse component, many bounces, R^N penalty, arrives late. Vikuiti raises total N_{pe} mainly through this population.

Key message

Total $N_{pe} \neq$ timing information carried by the *first* photons.

A reflector that doubles N_{pe} does not halve $\sigma(T_0)$.

The earliest photons drive timing; the rest give position and redundancy.



Estimator definition

Pool the first m photon-arrival times on each END face:

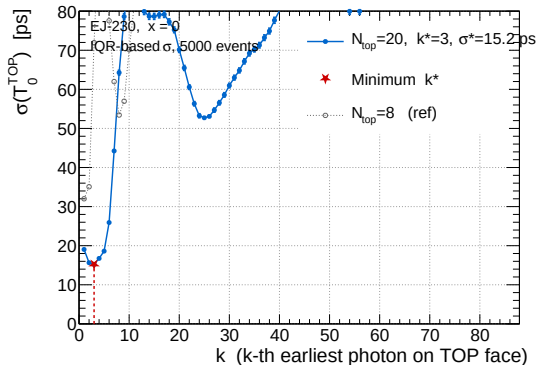
$$T_{\text{END}}(m) = \frac{1}{2} \left[\frac{1}{m} \sum_{i=1}^m t_L^{(i)} + \frac{1}{m} \sum_{i=1}^m t_R^{(i)} \right]$$

$t_L^{(i)}, t_R^{(i)}$: sorted arrival times on LEFT / RIGHT END face.

Two geometries

	m^*	σ_{END} (ps)
END-only	7	50.5
+TOP SiPMs	8	53.7

TOP SiPMs intercept early photons before they reach END $\Rightarrow +3$ ps.



Estimator definition

Pool *all* photon-arrival times from all N_{TOP} SiPMs on the TOP face. Sort the pooled list. Take the k -th earliest:

$$T_{\text{TOP}}(k) = t_{\text{TOP}}^{(k)}$$

$N_{\text{TOP}} = 20$: pool ~ 1340 photons/event. Optimal $k^* = 3$.

Why ~ 70 ps is not directly comparable

Old analysis: per-SiPM mean estimator (average first-photon over many SiPMs), EJ-230 TIR, $N = 70$ SiPMs $\rightarrow 70.2$ ps.

New analysis: pooled order statistic (all photons in one pool), EJ-230 Vikuiti, $N = 20$ sparse $\rightarrow 15.2$ ps.

The estimator advance is the primary driver of improvement.

BLUE = Best Linear Unbiased Estimator

Optimally combine two calibrated estimators of the same quantity T_0 :

$$T_{\text{BLUE}} = w_{\text{END}} T_{\text{END}} + w_{\text{TOP}} T_{\text{TOP}},$$

with $w_{\text{END}} + w_{\text{TOP}} = 1$. Weights minimise $\text{Var}(T_{\text{BLUE}})$ given the covariance matrix:

$$C = \begin{pmatrix} \sigma_{\text{END}}^2 & \text{Cov} \\ \text{Cov} & \sigma_{\text{TOP}}^2 \end{pmatrix}, \quad \text{Cov} = \rho \sigma_{\text{END}} \sigma_{\text{TOP}}.$$

BLUE result:

$$\vec{w} = \frac{C^{-1} \mathbf{1}}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad \sigma_{\text{BLUE}}^2 = \frac{1}{\mathbf{1}^T C^{-1} \mathbf{1}}.$$

For two estimators:

$$w_{\text{END}} = (\sigma_{\text{TOP}}^2 - \text{Cov}) / (\sigma_{\text{END}}^2 + \sigma_{\text{TOP}}^2 - 2 \text{Cov})$$

Physical interpretation

BLUE gives significant improvement only when both estimators carry *independent* information that the other lacks.

Applied here

Both T_{END} and T_{TOP} are functions of photons from the same muon event, so they are correlated: $\rho \approx 0.16$.

Weights computed event-by-event from the sample covariance matrix. Optimality is statistical, not per-event.

If END and TOP were independent ($\rho = 0$)

Inverse-variance combination:

$$\sigma_{\text{ind}} = \left(\frac{1}{\sigma_{\text{END}}^2} + \frac{1}{\sigma_{\text{TOP}}^2} \right)^{-1/2} \approx 14.6 \text{ ps}$$
$$w_{\text{END}}^{\text{ind}} \approx 0.074, \quad w_{\text{TOP}}^{\text{ind}} \approx 0.926.$$

Actual result ($\rho = 0.16$)

$$\sigma_{\text{BLUE}} = 15.21 \pm 0.26 \text{ ps}$$

$$w_{\text{END}} = 0.013, \quad w_{\text{TOP}} = 0.987$$

Interpretation

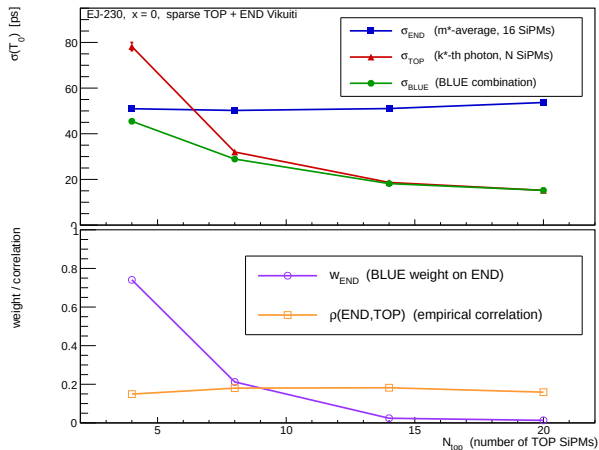
The correlation $\rho = 0.16$ pushes w_{END} from 0.074 (uncorrelated limit) to 0.013. Correlation reduces the independent information that END provides.

At $N_{\text{TOP}} = 20$, END contributes essentially no additional event-time information. $\sigma_{\text{BLUE}} \approx \sigma_{\text{TOP}}$ within statistical precision (0.01 ps difference, well inside 0.26 ps bootstrap uncertainty).

Correct statement

BLUE *does not improve* on TOP alone at $N_{\text{TOP}} = 20$. It is the statistically correct combination — it simply confirms that TOP already captures the available event-time information.

BLUE combination vs. number of TOP SiPMs



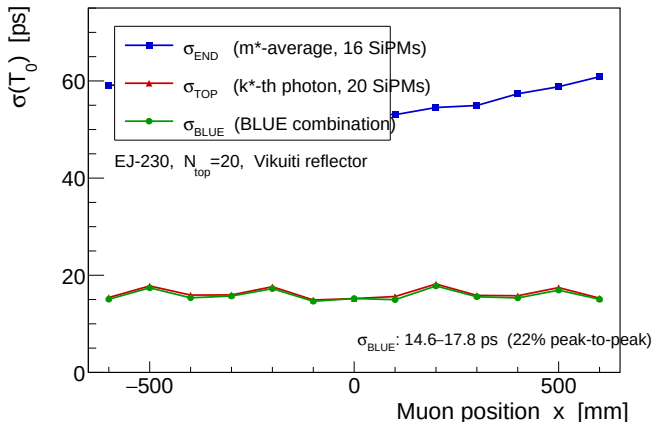
N	σ_{END} [ps]	σ_{TOP} [ps]	w_E	σ_{BLUE} [ps]
4	51.0	78.2	0.74	45.5
8	50.2	32.0	0.21	28.9
14	51.0	18.7	0.024	18.2
20	53.7	15.2	0.013	15.2

Message

BLUE genuinely combines information at low N . At high N , weights collapse onto TOP — correctly. This is a successful result, not a failure of BLUE.

$\rho \approx 0.16$ throughout (correlation does not depend on N_{TOP}).

$\sigma(T_0)$ vs. muon position — three estimators



EJ-230, $N_{\text{TOP}} = 20$, Vikuiti, optical limit.

Range

σ_{END} : 53–61 ps

σ_{TOP} : 14.9–18.2 ps

σ_{BLUE} : 14.6–17.8 ps

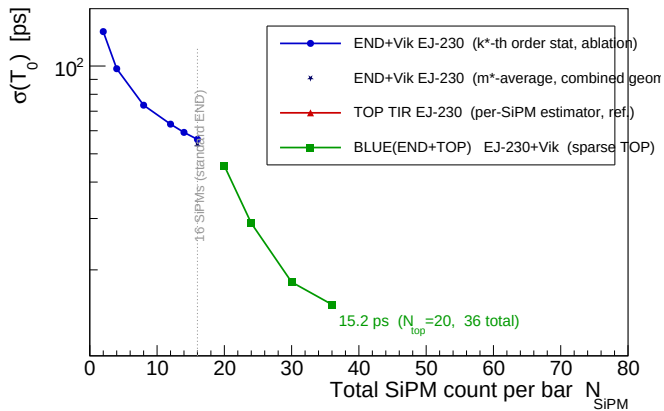
Peak-to-peak variation: **21%**. Not flat.

Physical origin

The variation tracks proximity to the nearest TOP SiPM, not λ_{att} . Three positions ($x \approx \pm 200, \pm 500$ mm) fall between SiPMs and show elevated $\sigma(T_0)$.

Implication: uniform timing response requires optimised sparse TOP placement, not simply more SiPMs.

Pareto frontier: timing resolution vs. channel count



Channel accounting (EJ-230):

Config	N_E	N_T	σ [ps]
END only	16	0	56
END m -avg	16	0	54
BLUE $N=4$	16	4	45.5
BLUE $N=8$	16	8	28.9
BLUE $N=14$	16	14	18.2
BLUE $N=20$	16	20	15.2

Adding 20 sparse TOP SiPMs to the baseline 16 END SiPMs (total 36 channels) reduces $\sigma(T_0)$ by a factor $\times 3.7$ vs. END alone.

TOP face

Strong event-time estimator. $\sigma_{\text{TOP}} = 15.2$ ps. Pooled order statistic on ~ 1340 photons/event. Provides T_0 to sub-20 ps precision across the bar.

END faces

Longitudinal position estimator. From the time difference:

$$\Delta t = t_R - t_L = -\frac{2x}{v_{\text{eff}}} \Rightarrow \sigma_x \approx 10.29 \text{ mm}$$

Also a redundant T_0 estimator, but with $3.5\times$ worse resolution than TOP.

Design question

The question is no longer simply:
How do we minimise σ_t ?

It is:

What is the minimum instrumentation that satisfies both the timing requirement and the longitudinal-position requirement?

With SiPM SPTR and front-end jitter included, the answer may differ from the optical-limit result here.

EJ-204 vs. EJ-230

At $N_{\text{TOP}} = 20$: $\sigma_{\text{BLUE}} = 15.14$ vs. 15.21 ps.
Difference $< 0.3\sigma$ — scintillator choice is irrelevant at this SiPM count.

1 — Napkin confirms geometry

TIR, propagation and repeated reflection predict the correct ordering of all three materials and close the photon budget to 20 %.

2 — Estimator matters as much as hardware

Pooled k-th order statistic (TOP, $N = 20$): 15.2 ps.
Per-SiPM mean estimator (old, $N = 70$): ~ 70 ps.
Same bar, same photons — $4.6\times$ difference from the estimator.

3 — END provides position; TOP provides time

At $N_{\text{TOP}} = 20$, BLUE weight on END $\approx 1.3\%$. END is nearly irrelevant for timing but essential for longitudinal position ($\sigma_x \approx 10.29$ mm).

4 — Optimisation is not finished

- Best tested point: $\sigma_{\text{BLUE}} = 15.2$ ps (36 SiPMs, optical limit).
- $\sigma(T_0)$ varies 21 % over the bar — TOP SiPM placement can be optimised for uniformity.
- SiPM SPTR and front-end jitter not included yet.
- EJ-204 and EJ-230 are statistically equivalent at $N_{\text{TOP}} = 20$.
- The Pareto front with jitter will determine the practical channel minimum.

Backup slides

Two-end readout

$$t_L = t_0 + (L/2 + x)/v_{\text{eff}} + \varepsilon_L$$

$$t_R = t_0 + (L/2 - x)/v_{\text{eff}} + \varepsilon_R$$

Sum: $T_0 = (t_L + t_R)/2$, x cancels

Diff: $\Delta t = -2x/v_{\text{eff}} \Rightarrow$ position

$$\sigma_x = (v_{\text{eff}}/2) \cdot \sigma_{\Delta t} \approx 10.29 \text{ mm}$$

Consistency check

$$v_{\text{eff}} = 184.6 \text{ mm/ns}, \sigma_{\Delta t} = 111.5 \text{ ps:}$$

$$\sigma_x \approx 10.29 \text{ mm.}$$

Position resolution is not independent information
— it is derived from the timing resolution.

Path-length spread limits timing

For large scintillators and fast emitters, the distribution of t_{prop} is the dominant limit (Knoll).

Why assume the first photon is optimal?

Current estimators:

- Leading-edge: use $t^{(k)}$ for $k = 1$
- Trimmed mean of first m photons (END)
- Pooled k -th order statistic (TOP)
- BLUE combination of END + TOP

Evidence that $k = 1$ is not always optimal

In an EJ-228 cylinder study, $\sigma(T_0)(k)$ is non-monotonic with minimum near $k \approx 8$. The number does not transfer; the principle does.

For EJ-230 END (this bar): $k^* = 2$ (k-scan) but $m^* = 7$ (m-average) gives 50.5 ps, better than any single-photon order statistic.

END k-scan caution

The k-scan minimum (56 ps at $k^* = 2$) is not the best END estimator. The m-average at $m^* = 7$ gives 50.5 ps (END-only geometry) and 53.7 ps in the combined geometry.

The Pareto frontier for END uses the k-scan values — it slightly understates the best achievable END timing.

Readout coupling	Acceptance toward one end	PE/end	Vik/TIR ratio
Index-matched	0.2742 (TIR-guided)	≈ 570.0 pe	≈ 1.2
Air gap (escape cone)	0.1129	≈ 234.6 pe	≈ 3
Reported run	—	≈ 1 PE	$\approx 10^3$

Escape cone cannot explain 10^3

The cone cuts both Vikuiti and TIR configurations by the same factor — it cannot generate the ratio. The napkin brackets the genuine ratio between 1 and 3.

Falsifiable prediction

Run END + TIR-only and END + Vikuiti under identical conditions. A ratio near 10^3 is a surface or RINDEX definition problem, not a physics result.

Specular — Vikuiti 3M ESR

Preserves angle of incidence. Keeps path-length dispersion small, favouring the earliest photons.

Diffuse — Teflon, TiO_2 , MgO

Lambert's law: outgoing direction independent of incoming. Every bounce is a fresh chance to redirect toward a readout window, at the cost of a broader t_{prop} distribution.

Napkin hypothesis (falsifiable)

At comparable reflectivity, diffuse scattering should raise collected light and broaden the arrival distribution. One campaign, one variable, prediction on record.

Knoll: diffuse reflectors generally collect more light, but that comparison is on yield, not timing.